VENKATA RAMI REDDY BOBBALA

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Summary

Java Backend Developer with 4 years of experience building scalable, production-grade distributed systems with Java and the Spring ecosystem. At Tata Consultancy Services, delivered event-driven microservices with Kafka, low-latency caching with Redis, and containerized deployments via Docker – reducing boilerplate code by 25% and improving API throughput by 40%. Skilled in database performance optimization, JWT-secured API design, and maintainable architecture.

Technical Skills

Languages: Java (8 / 11 / 17), SQL
Java & Backend: Spring Boot, Spring MVC, Spring Data JPA, Spring JDBC, Spring AOP, Spring AI, REST APIs, Lombok, OpenAI, LLM Tool Calling, ChatMemory
Microservices & Messaging: Microservices Architecture, Spring Cloud Gateway, API Gateway, Apache Kafka, gRPC, Event-Driven Architecture, Synchronous / Asynchronous Communication
Databases & Caching: PostgreSQL, MySQL, Indexing, Query Optimization, Redis, Spring Cache, Lettuce, Distributed Caching
Security: Spring Security, JWT, Authentication, Role-Based Authorization
System Design: Scalability, Load Balancing, Rate Limiting, Database Sharding
DevOps & Tooling: Docker, Docker Compose, Git, Maven, CI/CD Pipelines
Testing: JUnit, Mockito, Integration Testing, Testcontainers
Production Support: Log Analysis, Performance Profiling, Incident Investigation, Root Cause Analysis

Professional Experience

Tata Consultancy Services (TCS) Hyderabad, India
Java Backend Developer *Dec 2021 – Dec 2025*

- Architected and maintained scalable backend microservices using **Java 17**, **Spring Boot**, and **Spring MVC** for large-scale enterprise applications, supporting **10,000+ concurrent users** with 99.9% uptime.
- Designed and integrated **Apache Kafka** event-driven pipelines for asynchronous inter-service communication, reducing service coupling and improving system throughput by **40%** under peak load.
- Implemented **Redis** distributed caching using Spring Cache and Lettuce, cutting average API response times by **60%** for high-frequency read operations and eliminating redundant database hits.
- Containerized microservices using **Docker**, authoring Dockerfiles and Compose configurations that standardized dev/prod environments and reduced deployment setup time by **30%**.
- Drove database performance improvements across **PostgreSQL** and **MySQL** via strategic indexing and query refactoring through Spring JPA and Spring JDBC, reducing slow query count by **35%**.
- Engineered **JWT-based authentication** and role-based authorization using **Spring Security**, hardening API endpoints across production services.
- Led production incident investigations, diagnosing root causes through log analysis and performance profiling, delivering hotfixes with **sub-2-hour MTTR**.

Projects

AI Agents – Tool-Calling Agent Platform | *Spring AI, Spring Boot, OpenAI, Lombok* GitHub

- Built a conversational AI agent using **Spring AI** and **OpenAI** that performs LLM-driven tool orchestration, invoking the correct backend tool at runtime instead of hardcoded if/else routing.
- Implemented five **@Tool**-annotated tool sets across travel (flights, hotels, weather) and e-commerce support (orders, inventory), exposed through a single REST `/chat` endpoint.
- Added sliding-window **ChatMemory** to preserve conversational context across multi-turn interactions, enabling coherent handling of follow-up requests.

Patient Management System – Microservices Platform | *Java 17, Spring Boot, Kafka, gRPC, Docker* GitHub

- Designed and built a microservices-based Patient Management System with five independently deployable Spring Boot services – API Gateway, Auth, Patient, Billing, and Analytics – on an internal Docker network.
- Implemented **JWT-based authentication** and centralized routing via **Spring Cloud Gateway**, exposing a single public entry point while keeping downstream services internal-only.
- Integrated **gRPC** for synchronous Patient-to-Billing communication and **Apache Kafka** for asynchronous event streaming to a dedicated Analytics Service, with Docker-containerized services and isolated PostgreSQL databases per service.

Education

ACE Engineering College Hyderabad, India
Bachelor of Technology (B.Tech) in Electrical and Electronics Engineering *2017 – 2021*